

ME 643 | Aircraft and Rocket Propulsion

Aircraft Engine Design Project

Part 1: Request For Proposal

Selected Aircraft: For this project, our team (group 8) has been allotted the **Private Jet** category. So, we have selected the **Gulfstream G650**. The G650 is a premier ultra-long-range business jet produced by Gulfstream Aerospace.

Aircraft Specifications and Flight Conditions	
Seating capacity (Not for cargo aircraft)	13-16
Maximum payload (MP) (N)	$2,948\text{kg} \times 9.81 = 28,900\text{N}(\text{approx.})$
Maximum fuel capacity (N)	$20,049\text{kg} \times 9.81 = 196,860\text{N}$
Maximum takeoff weight (MTOW) (N)	45,178 kg
Maximum landing weight (N)	37,875 kg
Wing area (m ²)	119.2 m ²
Type of engine	Turbofan(2 × Rolls Royce BR 725- A1-12)
Number of engines	2
Maximum engine inlet diameter (m)	1.27
Cruise altitude (km)	15.5
Cruise Mach number	0.85
Range (km)	13000

Auxiliary data required for the project	
Aircraft manufacturing company	Gulfstream Aerospace
Departing airport	Sardar Vallabhbhai Patel International Airport
Arrival airport	Heathrow Airport

Auxiliary data required for the project	
Runway length (m)	3500 m
Single-engine climb gradient	2.4%
Ambient temperature	38°C
Airport altitude (km)	sea-level
Maximum lift coefficient, $C_{l,max}$	1.6
Turbine inlet temperature (T_{t4})	1500 K
Calorific value of jet fuel (h_{PR})	43 MJ/kg

Sources -

1. <https://www.aircharterservice.co.in/aircraft-guide/private/gulfstream-usa/gulfstream-g650>
2. <https://www.airplaneupdate.com/2019/06/gulfstream-g650.html>
3. <https://mjet.eu/fleets/g650/>
4. https://en.wikipedia.org/wiki/Gulfstream_G650/G700/G800

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